

ALIREZA MOHAMMADI



E-mail



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GitHub



Website



Google Scholar

EDUCATION

B.Sc. in Computer Engineering, Islamic Azad University

2020 - 2025

Kermanshah, Iran

CGPA: 16.54/20 (3.5/4)

RESEARCH INTERESTS

▪ AI Safety ▪ AI Alignment ▪ Explainable AI ▪ AI for science

ACADEMIC EXPERIENCE

Research Intern | ZEISS Lab @Medical University of Vienna, Austria (Remote)

Jan 2025 - Present

- Collaborated on designing and evaluating frequency-based explainability methods for neural networks.
- Co-authored manuscript with international team of researchers from Medical University of Vienna and ZEISS Lab.

Research Assistant | Islamic Azad University | Supervised by Dr.Parandin

2022 - 2025

▪ Contributed to the development and implementation of various ML models, including RNN, CNN, FNN and meta-learning frameworks.

Teaching Assistant | Islamic Azad University | Supervised by Dr.Habibi

2023

- TA of Computational Intelligence | ▪ TA of Natural Language Processing

As a Teaching Assistant, I conducted teaching sessions, led class discussions, and provided support in understanding complex concepts. I also assisted students with assignments, offered feedback, and guided them through coding and problem-solving exercises.

PUBLICATIONS

ACADEMIC JOURNALS

Citations 62 - h-index 4

1. **A Mohamadi**, A Yavari. "Survival at Any Cost? LLMs and the Choice Between Self-Preservation and Human Harm" (Preprint)
2. A Yavari, **A Mohamadi**, E Beydagh, R A. Leitzgeb. "FreqAttXpose: Frequency-Aware Model Parameter Explorer: A new attribution method for improving explainability" (Preprint)
3. **A Mohamadi**, F Parandin, P Karami. "Meta-Learning and Formula Optimization for All-Optical XOR, OR, and NOT Logic Gates: The ML-FOLD Method" (Under second-round review in EAAI) (**IF:8.0**)
4. F Parandin, **A Mohamadi**, P Karami. "Enhancing integrated optical circuits: optimizing all-optical NAND and NOR gates through deep learning and machine learning" Optical and Quantum Electronics (**IF:4.0**)
5. **A Mohamadi**, F Parandin, P Karami, S Olyaei. "Design and Optimization of Optical NAND and NOR Gates Using Photonic Crystals and the ML-FOLD Algorithm". Photonics (**IF:2.1**)
6. F Parandin, P Karami, **A Mohamadi**. "Machine Learning-Driven Optimization of Photonic Crystal Structures for Superior Optical NOR Gate Performance" Applied Optics (**IF:1.9**)

CONFERENCE PROCEEDINGS

7. **A Mohammadi**, H Ghahramani, SA Asghari, M Aminian. "Securing Healthcare with Deep Learning: A CNN-Based Model for medical IoT Threat Detection" 19th Iranian Conference on Intelligent Systems (**IEEE indexed**)
8. **A Mohammadi**, F Parandin, H Ghahramani. "Neural Network-Driven Optimization of Photonic Crystal-Based All-Optical NOT Gate Design" International Conference on Distributed Computing and High Performance Computing, 2024. (**IEEE indexed**)

9. F Parandin, **A Mohammadi**. "Enhancing the Performance of Photonic Crystal AND Gates with Machine Learning Optimization" International Conference on Distributed Computing and High Performance Computing,2024.  
(*IEEE indexed*)

 AWARDS & HONORS

- Received admission offer for the M.Sc. in Computer Engineering at NC State University (NC, USA) 2025
- One of [my papers](#) gained significant attention in the AI research community featured by several technology news outlets [\[1\]](#), [\[2\]](#) and was shared on social media platforms [\[3\]](#), [\[4\]](#) 2025
- Most cited paper, 19th Iranian Conference on Intelligent Systems. 2024
- Interviewed by Hamshahri newspaper and hispanTV as the Student Inventor  2016
- Selected idea for the 8th Student Festival Nanoscience and Nanotechnology  2015
- Recognized exceptional talent by National Organization for Development of Exceptional Talents 2014

 SKILLS

Programming	Python
Libraries	Scikit-learn, PyTorch, Auto-sklearn, TensorFlow, Matplotlib, NumPy, Pandas
Skills	Machine Learning, Data Analysis, Research Prowess, Optimization

 LANGUAGES

- TOEFL iBT: **95**/120 (R:25, L:23, S:24, W:23) CEFR C1 | Test Date: Jan 4, 2026
- Farsi: Native speaker

 PROJECTS

Securing Healthcare with Deep Learning: A CNN-Based Model for Medical IoT Threat Detection  ★ 24
Developed and implemented a CNN-based model for detecting threats in IoMT environments. The proposed model achieved a perfect accuracy of 0.99 across binary, categorical, and multiclass classification tasks, outperforming previous state-of-the-art methods.

DECIDE-SIM (*In collaboration with Med Uni of Vienna*)  ★ 6
DECIDE-SIM is a groundbreaking, open-source simulation framework designed to evaluate the ethical and cooperative behaviors of Large Language Model (LLM) agents in high-stakes survival scenarios. Our framework provides a systematic testbed to investigate how AI agents balance self-preservation, cooperation, and moral constraints when faced with resource scarcity and critical ethical dilemmas.